Project Proposal: Development of a Digital System for Managing Early Intervention Programs

Project information

Item	Details
Project title	Development of a Digital System for Managing Early Intervention Programs
Student	Yergesh Moldir Batyrbekkyzy
Supervisor	Basheyeva Zh.O.
Stage	Pre-defense 1: analysis and design
Repository	https://github.com/AAAman333/early-intervention-system
Task tracking	YouTrack project DIP (tasks DIP-1 to DIP-11)

Problem statement

Parents of children who need early intervention have no single place for the child's data, schedule, plan and progress, and must retell the child's history to every new specialist.

The need is large and growing. In Kazakhstan about one in four children aged 2–4 does not develop according to age norms (UNICEF, MICS 2024), and there are about 115 thousand children with disabilities (Ministry of Labour, 2025). State Development and Early Intervention Centres have opened at polyclinics since June 2025.

The project's own research confirms the problem:

- **Survey of 13 parents:** 62% learn the schedule through WhatsApp or Telegram; only 38% have one place where sessions, plan and progress are visible; 9 of 10 parents who changed specialists had to repeat the child's history; 31% get recommendations only when they ask.
- **3 interviews with parents:** schedule messages get lost or mixed up with several specialists; home recommendations are forgotten; progress over months is hard to see; both parents want access; all want to confirm access to the child's data themselves.
- **Existing systems** (CDC Milestone Tracker, Kinedu, DOQ.kz, Damumed) cover only parts of the process, and none lets the parent own the child's profile and share it with consent.

Details: docs/02-relevance.md, docs/03-literature-review.md, docs/04-existing-systems.md in the repository.

Target users

The system has four types of users; the child's data belongs to the parent.

User	Role in the system	Interface
Parent	Creates and fills in the child's account, finds a specialist, shares the child's data with consent and can revoke access, follows the plan, sessions, recommendations and progress	Mobile app (Flutter)
Child	User of the system; the account is created, filled in and opened by the parent; the child does not log in	Through the parent
Specialist	Speech therapist, psychologist, special educator, etc.: accepts requests, creates the individual plan, schedules sessions, records attendance, comments and home recommendations	Web application
Administrator	Verifies specialists, manages the catalogue of session types, sees the list of users and summary statistics	Web application

Aim and objectives

Aim: to develop a working prototype of an information system that helps parents notice developmental problems early, find a suitable specialist and share the child's data with consent, and helps specialists plan and track individual early intervention programs.

Objectives:

1. Analyse the domain, existing systems and literature; collect requirements from parents (done: survey of 13 parents, 3 interviews, 17 sources, 4 systems).
2. Compare technologies and choose a stack justified by the requirements .
3. Specify functional and non-functional requirements with priorities and acceptance criteria.
4. Design the architecture, data model and user interface .
5. Implement accounts and roles, child accounts managed by parents, specialist search, consent-based access, individual plans, scheduling, attendance, development screening and reports.
6. Test the key scenarios, run a usability test with 3–5 users and compare the system with the current way of working (Excel, paper, messengers).
7. Prepare documentation and a demonstration with test data.

Expected outcomes

The final result is a working prototype that passes one end-to-end scenario with demonstration data.

- Flutter mobile app for parents and React web application for specialists and administrators, sharing one Node.js backend and a PostgreSQL database.
- Documentation: requirements, architecture, diagrams, test plan and test results.
- Evaluation: all “Must” requirements pass their test cases; a usability test with 3–5 users; comparison with the current way of working.

Key acceptance scenario:

1. The parent creates the child’s account and completes the development screening.
2. The system recommends a type of specialist; the parent finds a verified specialist in their city and sends a request with access to the child’s data.
3. The specialist accepts, creates a plan (2 goals, 2 sessions a week for 4 weeks) and schedules 8 sessions without overlaps.
4. The specialist marks 6 sessions as held with comments and home recommendations, 1 as missed, 1 as cancelled.
5. The parent sees the history and recommendations; the report shows 8 planned, 6 held, 75% attendance.
6. The parent revokes access, and the specialist no longer sees the child.

Project scope

In scope	Out of scope
Registration and roles for parents, specialists and administrators; child accounts managed by parents	Processing real personal or medical data of children (only demonstration data)
Development screening based on CDC milestone checklists (not a diagnosis)	Clinical validation of the screening

In scope	Out of scope
Search of verified specialists by city and specialisation, ranked by rating and free time	Online payments and video sessions
Requests to specialists, consent-based data sharing, revoke, access log	Integration with state medical information systems (e.g. Damumed)
Individual plans with goals; catalogue of session types	Mobile app for specialists (they use the web application)
Calendar, recurring sessions, conflict check, attendance, comments, home recommendations	
Ratings after held sessions, reminders, reports on plan execution	

Proposed solution and technologie

The system connects the whole path in one place: screening, finding a specialist, sharing the parent-owned child profile with consent, and the plan with sessions, attendance and recommendations visible to both sides.

Layer	Technology	Why
Parent client	Flutter (Android, iOS)	92% of surveyed parents prefer a phone; one code base for both platforms
Specialist and admin client	React web application	Specialists work with tables, calendars and reports at a computer
Backend	Node.js + Express, REST API	One language (JavaScript) with the web client; one API for both clients
Database	PostgreSQL + Prisma	Strongly connected data and reports; database-level check of overlapping sessions
Authentication	JWT, bcrypt	Works for both web and mobile clients
Testing and tools	Jest, Supertest, flutter_test, Docker	Automated tests of core logic; one-command start

Layer	Technology Compose, GitHub, YouTrack	Why

Each choice was compared with 2–3 alternatives (docs/05-technology-comparison.md). UI mockups of the key scenarios are in docs/10-ui-prototype.md.

Supervisor's feedback

Item	Supervisor's comment
Problem and relevance	The project addresses a relevant and practically significant problem. The student conducted preliminary research, including a survey of 13 parents, three interviews, a review of 17 sources, and a comparison of four existing systems. During our consultations, we discussed the identified challenges in parent–specialist interaction and the need for a unified system for managing early intervention programs. I consider the relevance of the topic sufficiently justified for proceeding to the next stage.
Scope and engineering complexity	The project involves developing a mobile application for parents, a web application for specialists and administrators, and a shared backend and database. During our consultations, we reviewed the main functional modules and project boundaries. I recommended prioritizing the MVP functionality, focusing on individual intervention plans, session scheduling, attendance tracking, and parent–specialist interaction. The engineering complexity is appropriate for a Software Engineering diploma project.

Item	Supervisor’s comment
Requirements and architecture	The student has identified the main user groups, functional requirements, and the parent–specialist interaction scenario. During our consultation on October 23, we specifically discussed the child's role in the system. The recommendation was incorporated: the child's profile is created and managed by the parent. For the next stage, I recommended further detailing the architecture diagrams, database model, role-based access control, and main API interfaces.
Technology choice	The proposed technology stack includes Flutter for the mobile application, React for the web interface, Node.js/Express for the backend, and PostgreSQL for data storage. The student provided a rationale for the technology choices based on user requirements and the planned functionality. I consider the selected technologies appropriate for the project. The next stage should demonstrate their integration through the main user scenario.
Other comments and required changes	The student has demonstrated consistent progress in domain research, analysis of existing solutions, and preparation of project documentation. GitHub and YouTrack are used to organize the development process. Further supervision will focus on task completion and intermediate deliverables. I recommended finalizing requirement priorities, preparing a detailed data model, defining testing criteria, and implementing the main end-to-end

Item	Supervisor’s comment
	scenario using demonstration data. Developmental screening features should remain informational rather than diagnostic.
Decision (approved / approved with changes / revision needed)	Approved with changes. The presented work meets the expectations of the analysis and design stage. The student may proceed to prototype implementation while addressing the supervisor's recommendations. Further progress will be monitored through regular supervision meetings.

Previous feedback (consultation on 23.10.2026)**:** add the child to the users of the system. **Implemented:** the child is a user whose account is created, filled in and opened by the parent (requirements FR-37, FR-38; updated diagrams).

Supervisor: Basheyeva Zh.O.